

Qualification Pack



Mineral Processing Operator

Grinding/ Beneficiation/ Tailing

QP Code: MIN/Q4101

Version: 3.0

NSQF Level: 4

SCMS (Skill council of Mining Sector) || FIMI House, B-311, Okhla Industrial Area, Phase-I
New Delhi-110020 || email:scms@skillcms.in

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MIN/Q4101: Mineral Processing Operator

Brief Job Description

Mineral Processing Operator is engaged in mineral beneficiation in setting up crushing, screening, straining, washing, precipitating and other processes like leaching, floating and tailing management according to section in which posted and duties assigned and ensures proper working condition.

Personal Attributes

This job requires the individual to concentrate on the job at hand and complete it efficiently and effectively. The individual should be physically agile, strong, have good eye sight and not suffer from colour-blindness.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N4101: Pre-Operation checks of crusher and crusher sites](#)
2. [MIN/N4102: Operate conveyors and feeders](#)
3. [MIN/N4103: Operation and maintenance of the crusher](#)
4. [MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Options(Not mandatory):

Option 1: Grinding

1. [MIN/N4104: Grinding of crushed ore/mineral](#)

Option 2: Beneficiation

1. [MIN/N4105: Beneficiation and Mineral recovery](#)

Option 3: Tailing

1. [MIN/N4106: Tailing Management](#)

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Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
Country	India
NSQF Level	4
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8112.9900
Minimum Educational Qualification & Experience	12th grade Pass (or equivalent) OR 11th grade pass (with 1.5 year relevant experience in Mining Industry) OR 10th grade pass (with 3 years relevant experience in Mining Industry) OR Previous relevant Qualification of NSQF Level (3.0)(with 3 years relevant experience))
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	Not Applicable
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	06/05/2028
NSQF Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-04-MG-04063-2025-V2-SCMS
NQR Version	1.0

Remarks:

Total Notional Duration: 420 Hours; Additional 60 hours for Optional NOS= Domain Theory Duration: 100 Hours; Additional 20 hours for Optional NOS Theory Duration+ Domain Practical Duration: 290 Hours; Additional 40 hours for Optional NOS Practical Duration+ Employability Skills Duration: 30 Hours

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MIN/N4101: Pre-Operation checks of crusher and crusher sites

Description

This OS unit is about pre-operation check of crusher and crusher sites such as feeder, conveyor, VSI etc. before starting the crusher for operation.

Scope

The scope covers the following :

- Pre-Operation check of crusher and crusher sites

Elements and Performance Criteria

Pre-Operation check of crusher and crusher sites

To be competent, the user/individual on the job must be able to:

- PC1.** check the electric panel and electric supply before starting the crusher
- PC2.** check availability of raw materials at the site
- PC3.** inspect the moveable parts in crushers like conveyor belts, rollers, stringers, etc. for wear and tear
- PC4.** check Vertical Shaft Impactor (VSI) crusher, tips, liners, etc.
- PC5.** coordinate with other vehicle operators for procurement of the raw material as per standards of the organisation
- PC6.** inspect the suitable fire extinguishers and automatically operated fire suppression system and devices including automatic fire detection system at site
- PC7.** interpret the layout and identify components of belt conveyor system including its principal components like, trough, conveyor belt, clips, roller, guards, tensioners, feed and discharge chutes, head, tail and take-up pulleys, bearings, controls such as switches, trip cords, Inter locks, alarms, etc.
- PC8.** discuss the operations of mechanical and electrical driving units including safety devices
- PC9.** conduct conveying of the ore to the processing unit considering parameters like feed/ material rate, alignment of conveyor belts, spillage amount and rate, conveyor jams etc.
- PC10.** take corrective actions against any safety hazards
- PC11.** conduct operational checks on areas of potential issues like unusual noises/ smells, blockages and obstruction, lubrication of drive head, tail end and rollers , leaks, ensure magnets are operative
- PC12.** take corrective actions on any unusual issues observed during operational checks
- PC13.** check principal components of hoppers and bins, like grizzly rails, chutes, guardrails etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same

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- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** the hierarchy for reporting identified problems
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process
- KU6.** different types of mines and detail of the mine one is working in
- KU7.** benching in quarries, dressing of overhangs, undercuts, fencing
- KU8.** importance of first aid and hygiene
- KU9.** code of practice in specific areas of the mine
- KU10.** standing orders in force at the mine
- KU11.** importance of safety in the vicinity of machinery
- KU12.** about shot-firing / blasting related safety regulations including taking shelter during blasting
- KU13.** duties of workmen under the Mines Act-1952
- KU14.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU15.** the outcome of violation of safety procedures
- KU16.** emergency response /disaster management plan prepared by the organization
- KU17.** different types of ore processing processes and associated equipment
- KU18.** mechanism of transmission of power
- KU19.** types of raw materials and its physical qualities
- KU20.** basic electrical functioning and repairs
- KU21.** about Vertical Shaft Impactor (VSI) crusher, tips, liners, etc.
- KU22.** various moveable part of crusher such as conveyor belts, rollers, stringers, etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** discuss task lists, schedules and activities
- GS5.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS6.** make decisions pertaining to the concerned area of work
- GS7.** plan and organize the work order and tasks
- GS8.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS9.** detect problems in day to day tasks
- GS10.** discuss possible solution with the supervisor for problem solving
- GS11.** make decisions in emergency conditions
- GS12.** follow instructions and work on areas of improvement identified
- GS13.** complete the assigned tasks timely
- GS14.** use reasoning skills to identify and resolve basic problems



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- GS15.** analyze and detect any potential problems which could arise during operation
- GS16.** respect persons with disabilities
- GS17.** explain the importance of gender-sensitization at work site
- GS18.** state basic laws, acts and provisions defined for Persons with Disability (PwD) by the statutory bodies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre-Operation check of crusher and crusher sites</i>	30	50	-	20
PC1. check the electric panel and electric supply before starting the crusher	3	3	-	1
PC2. check availability of raw materials at the site	3	3	-	1
PC3. inspect the moveable parts in crushers like conveyor belts, rollers, stringers, etc. for wear and tear	2	4	-	1
PC4. check Vertical Shaft Impactor (VSI) crusher, tips, liners, etc.	2	4	-	1
PC5. coordinate with other vehicle operators for procurement of the raw material as per standards of the organisation	3	4	-	1
PC6. inspect the suitable fire extinguishers and automatically operated fire suppression system and devices including automatic fire detection system at site	2	4	-	1
PC7. interpret the layout and identify components of belt conveyor system including its principal components like, trough, conveyor belt, clips, roller, guards, tensioners, feed and discharge chutes, head, tail and take-up pulleys, bearings, controls such as switches, trip cords, Inter locks, alarms, etc.	2	4	-	2
PC8. discuss the operations of mechanical and electrical driving units including safety devices	3	4	-	2
PC9. conduct conveying of the ore to the processing unit considering parameters like feed/ material rate, alignment of conveyor belts, spillage amount and rate, conveyor jams etc.	2	4	-	2
PC10. take corrective actions against any safety hazards	2	4	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. conduct operational checks on areas of potential issues like unusual noises/ smells, blockages and obstruction, lubrication of drive head, tail end and rollers , leaks, ensure magnets are operative	2	4	-	2
PC12. take corrective actions on any unusual issues observed during operational checks	2	4	-	2
PC13. check principal components of hoppers and bins, like grizzly rails, chutes, guardrails etc.	2	4	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4101
NOS Name	Pre-Operation checks of crusher and crusher sites
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	4
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

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MIN/N4102: Operate conveyors and feeders

Description

This OS unit is about operate conveyors and feeders.

Scope

The scope covers the following :

- Operate the feeders
- Operate the conveyors

Elements and Performance Criteria

Operate the feeders

To be competent, the user/individual on the job must be able to:

- PC1.** interpret the layout and recognize the components of feeders including chains, pans, bearings, rollers, shafts, feed/discharge chute, drive mechanisms, lubrication systems, controls, drive belts, hangers/cables, etc.
- PC2.** conduct feeding of the ore to the crusher considering parameters like feed/ material rate, tilt, jams etc.
- PC3.** take corrective actions against any safety hazards like spillage etc. and issues like unusual noises/ smells, blockages and obstruction, lubrication, leaks, etc.

Operate the conveyors

To be competent, the user/individual on the job must be able to:

- PC4.** follow the safe code of practice for erection, installation, operation, repairs, maintenance, dismantling of plant and ancillary equipment and for the prevention of accident and to provide safety , health, convenience and discipline of the workers
- PC5.** check the suitable fire extinguishers and automatically operated fire suppression system and devices including fire detection system
- PC6.** interpret the layout and recognize the components of a belt conveyor system including its principal components like trough, conveyor belt, clips, roller, guards, tensioners, feed and discharge chutes, head, tail and take-up pulleys, bearings, controls such as switches, trip cords, Interlocks, alarms, etc. and operations of mechanical and electrical driving units including safety devices
- PC7.** conduct conveying of the ore to the processing unit considering parameters like feed/ material rate, alignment of conveyor belts, spillage amount and rate, conveyor jams etc.
- PC8.** take corrective actions against any safety hazards
- PC9.** conduct operational checks on areas of potential issues like unusual noises/ smells, blockages and obstruction, lubrication of drive head, tail end and rollers, leaks, ensure magnets are operative and take corrective actions if required
- PC10.** ensure principal components of hoppers and bins, like grizzly rails, chutes, guardrails, etc. are deployed to control stockpile levels, stability of stockpiles, clearing of the obstruction etc.

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** the hierarchy for reporting identified problems
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process
- KU6.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to mineral processing operations
- KU7.** different types of mines and detail of the mine one is working in
- KU8.** benching in quarries, dressing of overhangs, undercuts, fencing
- KU9.** importance of first aid and hygiene
- KU10.** code of practice in specific areas of the mine
- KU11.** standing orders in force at the mine
- KU12.** importance of safety in the vicinity of machinery
- KU13.** about shot-firing / blasting related safety regulations including taking shelter during blasting
- KU14.** duties of workmen under the Mines Act-1952
- KU15.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU16.** the outcome of violation of safety procedures
- KU17.** emergency response /disaster management plan prepared by the organization
- KU18.** different types of mineral processing processes and associated equipment
- KU19.** mechanism of transmission of power
- KU20.** proper signals and their correct interpretation
- KU21.** use of warning bells and application of safety devices
- KU22.** operation of different levers and operating switches and loading of mineral
- KU23.** different types of conveyors like belt conveyors, drag conveyors, pneumatic conveyors, airslid conveyors, screw conveyors, bucket elevators and pan conveyors
- KU24.** different types of conveyors principal components, such as screw, screw casing/trough, scrapers, conveyor belt, splices or clips, drive mechanisms, roller chains/belts, buckets, guards, tensioners, feed and discharge chutes, head, tail and take-up pulleys, idlers, bearings, braking system, controls (such as switches, trip cords, side travel switches, level probes, zero speed switches, inter locks, crossovers, audible alarms, magnet, metal detectors, flush water, lubricating systems, weight metre
- KU25.** different types of principal components of hoppers and bins, such as: grizzly rails, chutes, guardrails, entry doors, level probes, covers etc.
- KU26.** different types of feeders which include but are not limited to: pan feeder, screw feeder, belt feeder, pneumatic feeder, tube feeder, vibratory feeder, tripper, apron feeder, rotary valve feeder, drag feeder etc.
- KU27.** different feeders principal components, such as: chains, pans, bearings, (support, hangers, pillow block bearings), sprockets, rollers, roller mounts, return rollers, shafts, centre support rail, feed/discharge chute, drive mechanisms, gear reducer, lubrication systems, guards, skirt boards, controls, scraper, fire extinguishing equipment, drive belts, hangers/cables,etc.
- KU28.** specific safety instructions for conveying systems (pull chord, label switch, fire alarm etc.)

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- KU29.** use of hand drills, guards etc
- KU30.** belt mechanisms/systems (belt fastening, belt jointing etc.)
- KU31.** hazards and safety aspects involved in ore processing activities, and usage of relevant PPEs

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** enter the information in online Enterprise resource planning (ERP) systems under guidance of the supervisor
- GS5.** discuss task lists, schedules and activities
- GS6.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS7.** make decisions pertaining to the concerned area of work
- GS8.** plan and organize the work order and tasks
- GS9.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS10.** detect problems in day to day tasks
- GS11.** discuss possible solution with the supervisor for problem solving
- GS12.** make decisions in emergency conditions
- GS13.** follow instructions and work on areas of improvement identified
- GS14.** complete the assigned tasks timely
- GS15.** use reasoning skills to identify and resolve basic problems
- GS16.** analyze and detect any potential problems which could arise during operation
- GS17.** respect persons with disabilities
- GS18.** explain the importance of gender-sensitization at work site
- GS19.** state basic laws, acts and provisions defined for Persons with Disability (PwD) by the statutory bodies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate the feeders</i>	9	15	-	6
PC1. interpret the layout and recognize the components of feeders including chains, pans, bearings, rollers, shafts, feed/discharge chute, drive mechanisms, lubrication systems, controls, drive belts, hangers/cables, etc.	3	5	-	2
PC2. conduct feeding of the ore to the crusher considering parameters like feed/ material rate, tilt, jams etc.	3	5	-	2
PC3. take corrective actions against any safety hazards like spillage etc. and issues like unusual noises/ smells, blockages and obstruction, lubrication, leaks, etc.	3	5	-	2
<i>Operate the conveyors</i>	21	35	-	14
PC4. follow the safe code of practice for erection, installation, operation, repairs, maintenance, dismantling of plant and ancillary equipment and for the prevention of accident and to provide safety , health, convenience and discipline of the workers	3	5	-	2
PC5. check the suitable fire extinguishers and automatically operated fire suppression system and devices including fire detection system	3	5	-	2
PC6. interpret the layout and recognize the components of a belt conveyor system including its principal components like trough, conveyor belt, clips, roller, guards, tensioners, feed and discharge chutes, head, tail and take-up pulleys, bearings, controls such as switches, trip cords, Interlocks, alarms, etc. and operations of mechanical and electrical driving units including safety devices	3	5	-	2
PC7. conduct conveying of the ore to the processing unit considering parameters like feed/ material rate, alignment of conveyor belts, spillage amount and rate, conveyor jams etc.	3	5	-	2
PC8. take corrective actions against any safety hazards	3	5	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. conduct operational checks on areas of potential issues like unusual noises/ smells, blockages and obstruction, lubrication of drive head, tail end and rollers, leaks, ensure magnets are operative and take corrective actions if required	3	5	-	2
PC10. ensure principal components of hoppers and bins, like grizzly rails, chutes, guardrails, etc. are deployed to control stockpile levels, stability of stockpiles, clearing of the obstruction etc.	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4102
NOS Name	Operate conveyors and feeders
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	4
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

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MIN/N4103: Operation and maintenance of the crusher

Description

This OS unit is about operating crusher and conduct maintenance of the crusher.

Scope

The scope covers the following :

- Operation of the crusher
- Operate the crushing screen
- Maintenance and trouble shooting of the crusher

Elements and Performance Criteria

Operation of the crusher

To be competent, the user/individual on the job must be able to:

- PC1.** start and stop the crusher appropriately
- PC2.** discuss various types of crusher and their crushing principle
- PC3.** identify the important components of a jaw crusher, cone crusher and etc.
- PC4.** discuss the various methods of comminution
- PC5.** check different types of crushing principal components, such as gates, chutes and gaps, feeder, conveyor, dust collectors, water sprays etc.
- PC6.** identify the components towards crushing of the material like drive mechanism, pulleys, crushers, etc.
- PC7.** conduct crushing operations by controlling the feed/material by adjusting opening of gates, chutes and gaps, adjusting speed of feeder, adjusting speed of conveyor, controlling dust by using dust collectors, water sprays, remove and clean excess spillage, etc.
- PC8.** use and identify signals from assistant to start or stop the equipment
- PC9.** check the continuity of flow, safety, and efficient operation as per the instructions
- PC10.** check for proper graded flow of materials into the crushers and control the speed and flow of material through conveyors
- PC11.** test samples materials or products to ensure compliance with Indian standard(IS) specifications
- PC12.** conduct operational checks on areas of potential issues like unusual noises/ smells, blockages and obstruction, leaks etc. and take corrective actions if required
- PC13.** identify the areas of blockage and clear the crushing blockages periodically
- PC14.** coordinate with the vehicle operators for collection of the output as per instructions from operator/ supervisor safety during operation

Operate the crushing screen

To be competent, the user/individual on the job must be able to:

- PC15.** recognize the process requirements for separating feed/material and working around screens, including reporting, control of dust, load limitations, control of feed/material flow rate etc.

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PC16. conduct operational checks on areas of potential issues like unusual noises/ smells, blockages and obstruction, leaks etc.

Maintenance and trouble shooting of the crusher

To be competent, the user/individual on the job must be able to:

PC17. replace crusher component/wear part as per requirement or schedule

PC18. check indicators that signal need for replacement

PC19. replenish coolants, lubricants, fluids and screeners as and when required

PC20. follow the safe code of practice for erection, installation, operation, repairs, maintenance, dismantling of plant and ancillary equipment and for the prevention of accident and to provide safety, health, convenience and discipline of the persons so employed

PC21. identify missing or defective components or controls and replace them with genuine OEM recommended components

PC22. check oil, fuel tanks for leaks and take necessary actions as per the operational manual

PC23. lubricate all the moving parts at regular intervals

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. job-specific documents e.g. daily maintenance checklist and importance of the same

KU2. risk and impact of not following defined procedures/work instructions

KU3. the hierarchy for reporting identified problems

KU4. cost of equipment and loss for the company that results from damage of equipment

KU5. implications of delays in the process

KU6. safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to mineral processing operations

KU7. different types of mines and detail of the mine one is working in

KU8. benching in quarries, dressing of overhangs, undercuts, fencing

KU9. importance of first aid and hygiene

KU10. code of practice in specific areas of the mine

KU11. standing orders in force at the mine

KU12. importance of safety in the vicinity of machinery

KU13. about shot-firing / blasting related safety regulations including taking shelter during blasting

KU14. duties of workmen under the Mines Act 1952

KU15. provision of compensation and working hours, leaves, etc. as per Mines Act 1952

KU16. the outcome of violation of safety procedures

KU17. emergency response /disaster management plan prepared by the organization

KU18. different types of crusher and their importance

KU19. different types of crushing principal components, such as gates, chutes and gaps, feeder, conveyor, dust collectors, water sprays etc.

KU20. different types of screening equipment like drive mechanism, balance wheel, rocker arms, conveyor belts, guards, etc.

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KU21. hazards and safety aspects involved in ore processing activities and usage of relevant PPEs

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** discuss task lists, schedules and activities
- GS5.** enter the information in online Enterprise resource planning (ERP) systems under guidance of the supervisor
- GS6.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS7.** make decisions pertaining to the concerned area of work
- GS8.** plan and organize the work order and tasks
- GS9.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS10.** detect problems in day to day tasks
- GS11.** discuss possible solution with the supervisor for problem solving
- GS12.** make decisions in emergency conditions
- GS13.** follow instructions and work on areas of improvement identified
- GS14.** complete the assigned tasks timely
- GS15.** use reasoning skills to identify and resolve basic problems
- GS16.** analyze and detect any potential problems which could arise during operation
- GS17.** respect persons with disabilities
- GS18.** explain the importance of gender-sensitization at work site
- GS19.** state basic laws, acts and provisions defined for Persons with Disability (PwD) by the statutory bodies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operation of the crusher</i>	21	32	-	11
PC1. start and stop the crusher appropriately	2	2	-	1
PC2. discuss various types of crusher and their crushing principle	2	3	-	-
PC3. identify the important components of a jaw crusher, cone crusher and etc.	2	2	-	1
PC4. discuss the various methods of comminution	2	3	-	-
PC5. check different types of crushing principal components, such as gates, chutes and gaps, feeder, conveyor, dust collectors, water sprays etc.	1	3	-	1
PC6. identify the components towards crushing of the material like drive mechanism, pulleys, crushers, etc.	2	2	-	1
PC7. conduct crushing operations by controlling the feed/material by adjusting opening of gates, chutes and gaps, adjusting speed of feeder, adjusting speed of conveyor, controlling dust by using dust collectors, water sprays, remove and clean excess spillage, etc.	2	3	-	-
PC8. use and identify signals from assistant to start or stop the equipment	2	2	-	1
PC9. check the continuity of flow, safety, and efficient operation as per the instructions	1	2	-	1
PC10. check for proper graded flow of materials into the crushers and control the speed and flow of material through conveyors	1	2	-	1
PC11. test samples materials or products to ensure compliance with Indian standard(IS) specifications	1	2	-	1
PC12. conduct operational checks on areas of potential issues like unusual noises/ smells, blockages and obstruction, leaks etc. and take corrective actions if required	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. identify the areas of blockage and clear the crushing blockages periodically	1	2	-	1
PC14. coordinate with the vehicle operators for collection of the output as per instructions from operator/ supervisor safety during operation	1	2	-	1
<i>Operate the crushing screen</i>	2	4	-	2
PC15. recognize the process requirements for separating feed/material and working around screens, including reporting, control of dust, load limitations, control of feed/material flow rate etc.	1	2	-	1
PC16. conduct operational checks on areas of potential issues like unusual noises/ smells, blockages and obstruction, leaks etc.	1	2	-	1
<i>Maintenance and trouble shooting of the crusher</i>	7	14	-	7
PC17. replace crusher component/wear part as per requirement or schedule	1	2	-	1
PC18. check indicators that signal need for replacement	1	2	-	1
PC19. replenish coolants, lubricants, fluids and screeners as and when required	1	2	-	1
PC20. follow the safe code of practice for erection, installation, operation, repairs, maintenance, dismantling of plant and ancillary equipment and for the prevention of accident and to provide safety , health, convenience and discipline of the persons so employed	1	2	-	1
PC21. identify missing or defective components or controls and replace them with genuine OEM recommended components	1	2	-	1
PC22. check oil, fuel tanks for leaks and take necessary actions as per the operational manual	1	2	-	1
PC23. lubricate all the moving parts at regular intervals	1	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4103
NOS Name	Operation and maintenance of the crusher
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	3
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines

Description

This unit is about adhering to health, safety and environmental guidelines at the Open Cast Mines while executing various tasks, maintenance and operations.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow Environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** comply with safety, health, and security-related regulations/guidelines at the opencast mine
- PC2.** follow the safety instructions given by the workman's inspector
- PC3.** follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.
- PC4.** take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations
- PC5.** follow appropriate Safe Operating Procedure (SOP) while dealing with explosives
- PC6.** respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility
- PC7.** provide first aid to an injured person
- PC8.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC9.** use appropriate PPE as per the requirement
- PC10.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC11.** disinfect the machine/tools before and after work/task
- PC12.** maintain hygiene at the work site
- PC13.** report any symptoms of illness to the shift-incharge

Follow Environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC14.** identify the environmental impact of related opencast mining operations
- PC15.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC16.** ensure not to mix topsoil with waste in day to day tasks
- PC17.** ensure that HEMM is washed at the designated location

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- PC18.** ensure the productivity of the machine for material/fuel conservation
- PC19.** follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)
- PC20.** assist supervisor for reducing environmental impact caused due to related mining operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** about various environmental awareness program related to mining, organized by the various government bodies/company
- KU2.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU3.** how to identify six-directional hazards
- KU4.** basic mining terminologies and definitions
- KU5.** importance of blind spot survey/training
- KU6.** means of access and egress from the mines, location of workshop, haul roads and working face including dump yards
- KU7.** duties of workers under The Mines act-1952
- KU8.** working hours and accident compensation as per The Mines act-1952
- KU9.** about precautions to be taken when handling heavy equipment
- KU10.** various problems/incidents likely to occur
- KU11.** hierarchy of the reporting
- KU12.** machine operation, condition of the machine and worksite
- KU13.** proper documents specific to the machine
- KU14.** role of workmen inspector, safety committee and internal safety organization
- KU15.** the process of top soil removal and management
- KU16.** mine sump and pumping system of the mines
- KU17.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU18.** common sources of pollution in the mines and ways to minimize it
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, gas detector, safety lamp, self-contained breathing apparatus, gum boots, ear plugs, face mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization
- KU22.** signages, mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** basic personal and workplace hygiene
- KU25.** importance of sensitization towards different genders and PWD (Persons with Disabilities)

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS4.** plan and organize the work order and jobs
- GS5.** use reasoning skills to identify and resolve fundamental problems
- GS6.** complete the assigned tasks timely

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	20	34	-	15
PC1. comply with safety, health, and security-related regulations/guidelines at the opencast mine	2	2	-	1
PC2. follow the safety instructions given by the workman's inspector	1	3	-	1
PC3. follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.	3	3	-	1
PC4. take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations	1	3	-	1
PC5. follow appropriate Safe Operating Procedure (SOP) while dealing with explosives	2	3	-	2
PC6. respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility	2	3	-	1
PC7. provide first aid to an injured person	2	3	-	1
PC8. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	3	-	1
PC9. use appropriate PPE as per the requirement	2	4	-	2
PC10. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	2	-	1
PC11. disinfect the machine/tools before and after work/task	1	2	-	1
PC12. maintain hygiene at the work site	1	1	-	1
PC13. report any symptoms of illness to the shift-incharge	1	2	-	1
<i>Follow Environmental guidelines</i>	10	16	-	5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. identify the environmental impact of related opencast mining operations	2	2	-	1
PC15. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	2	-	1
PC16. ensure not to mix topsoil with waste in day to day tasks	2	2	-	1
PC17. ensure that HEMM is washed at the designated location	2	2	-	1
PC18. ensure the productivity of the machine for material/fuel conservation	1	3	-	-
PC19. follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)	1	3	-	1
PC20. assist supervisor for reducing environmental impact caused due to related mining operations	1	2	-	-
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1703
NOS Name	Follow Health, Safety, and Environmental Guidelines for opencast mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, , Exploration, Mineral Estimation, Planning, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Opencast, Specialized Operations, Electrical Services, Mechanical Services, Instrumentation and Control Systems, Ore Processing
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQF Clearance Date	11/03/2026

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/09/2026
NSQC Clearance Date	12/03/2026

Qualification Pack

MIN/N4104: Grinding of crushed ore/mineral

Description

This OS unit is about grinding the crushed ore/mineral.

Scope

The scope covers the following :

- Operate the grind mill

Elements and Performance Criteria

Operate the grind mill

To be competent, the user/individual on the job must be able to:

- PC1.** interpret the layout and recognize components of grinding mills including its principal components like drive motor, motor cooling fan, drive shaft and bearings, clutch, pinion gear, feed chute, shell, liners, lifters and bolts, ring/bull gear, trunnion and bearings, trommel screen, lubrication system, grinding media, controls, like main disconnect, Interlocks, stop/start switches (remote/local), selector switch remote/local), cooling system(s), sound boxes etc.
- PC2.** conduct grinding/ classifier operations considering parameters like load limitations, charging grinding media as required, control of dust, control of feed/material flow rate, control and maintenance of densities, sampling and testing, proper grind, reagent addition etc.
- PC3.** ensure that the control switches are operative and free of build-up and/or obstructions
- PC4.** ensure the lubrication system(s) are at required levels
- PC5.** ensure that the cooling system(s) are operative and at required levels
- PC6.** ensure that the feed and discharge points are free of obstructions and blockages
- PC7.** guard all identified hazards using rope/ bafflers and/or signs, clean up spills and leaks, etc.
- PC8.** inspect classifier and components, such as feed gate, discharge point, trash screen, guards, cyclone, feed lines, launders, skirting, upstream and downstream equipment and related systems, rollers, spirals
- PC9.** ensure wearing of fall arrest system, control required quantity /quality of discharge by sampling and testing as required, adjusting water flow, adjusting feed rate, control dust by using: dust collectors, water sprays, ensure minimum spillage, etc.
- PC10.** control density (percent solid, SG) by sampling
- PC11.** conduct operational checks on areas of potential issues like unusual noises/ smells, blockages and obstruction; Take corrective actions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same

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- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** the hierarchy for reporting identified problems
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process
- KU6.** handover and takeover procedures of the mineral processing operations according to company's SOP
- KU7.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to mineral processing operations
- KU8.** different types of mines and detail of the mine one is working in
- KU9.** benching in quarries, dressing of overhangs, undercuts, fencing
- KU10.** importance of first aid and hygiene
- KU11.** code of practice in specific areas of the mine
- KU12.** standing orders in force at the mine
- KU13.** importance of safety in the vicinity of machinery
- KU14.** about shot-firing / blasting related safety regulations including taking shelter during blasting
- KU15.** duties of workmen under the Mines Act-1952
- KU16.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU17.** the outcome of violation of safety procedures
- KU18.** emergency response /disaster management plan prepared by the organization
- KU19.** different types of ore processing processes and associated equipment
- KU20.** specific safety instructions for conveying systems (pull chord, label switch, fire alarm etc.)
- KU21.** different types of mills include: ball mill, pebble mill, autogenous mill/scrubber, semi-autogenous mill, rod mill, regrind mill
- KU22.** different components of mills including its principal components like drive motor, motor cooling fan, drive shaft and bearings, clutch, pinion gear, feed chute, shell, liners, lifters and bolts, ring/bull gear, trunnion and bearings, trommel screen, lubrication system, grinding media, controls, like Interlocks, stop/start switches (remote/local), selector switch (remote/local), cooling system(s), etc.
- KU23.** use of hand drills, guards etc.
- KU24.** belt mechanisms/systems (belt fastening, belt jointing etc.)
- KU25.** hazards and safety aspects involved in ore processing activities, and usage of relevant PPEs

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** enter the information in online Enterprise resource planning (ERP) systems under guidance of the supervisor
- GS5.** discuss task lists, schedules and activities



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- GS6.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS7.** make decisions pertaining to the concerned area of work
- GS8.** plan and organize the work order and tasks
- GS9.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS10.** detect problems in day to day tasks
- GS11.** discuss possible solution with the supervisor for problem solving
- GS12.** make decisions in emergency conditions
- GS13.** follow instructions and work on areas of improvement identified
- GS14.** complete the assigned tasks timely
- GS15.** use reasoning skills to identify and resolve basic problems
- GS16.** analyze and detect any potential problems which could arise during operation
- GS17.** respect persons with disabilities
- GS18.** explain the importance of gender-sensitization at work site
- GS19.** state basic laws, acts and provisions defined for Persons with Disability (PwD) by the statutory bodies

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate the grind mill</i>	30	50	-	20
PC1. interpret the layout and recognize components of grinding mills including its principal components like drive motor, motor cooling fan, drive shaft and bearings, clutch, pinion gear, feed chute, shell, liners, lifters and bolts, ring/bull gear, trunnion and bearings, trommel screen, lubrication system, grinding media, controls, like main disconnect, Interlocks, stop/start switches (remote/local), selector switch remote/local), cooling system(s), sound boxes etc.	3	4	-	1
PC2. conduct grinding/ classifier operations considering parameters like load limitations, charging grinding media as required, control of dust, control of feed/material flow rate, control and maintenance of densities, sampling and testing, proper grind, reagent addition etc.	3	5	-	2
PC3. ensure that the control switches are operative and free of build-up and/or obstructions	3	4	-	1
PC4. ensure the lubrication system(s) are at required levels	3	4	-	2
PC5. ensure that the cooling system(s) are operative and at required levels	3	4	-	2
PC6. ensure that the feed and discharge points are free of obstructions and blockages	2	5	-	2
PC7. guard all identified hazards using rope/ bafflers and/or signs, clean up spills and leaks, etc.	2	5	-	2
PC8. inspect classifier and components, such as feed gate, discharge point, trash screen, guards, cyclone, feed lines, launders, skirting, upstream and downstream equipment and related systems, rollers, Spirals	3	4	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. ensure wearing of fall arrest system, control required quantity /quality of discharge by sampling and testing as required, adjusting water flow, adjusting feed rate, control dust by using: dust collectors, water sprays, ensure minimum spillage, etc.	3	5	-	2
PC10. control density (percent solid, SG) by sampling	2	5	-	2
PC11. conduct operational checks on areas of potential issues like unusual noises/ smells, blockages and obstruction; Take corrective actions	3	5	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4104
NOS Name	Grinding of crushed ore/mineral
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	2
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

MIN/N4105: Beneficiation and Mineral recovery

Description

This OS unit is about recognizing the right beneficiation process to be executed, what reagents will be used, what equipment/ machines will be used and what is the required output considering the standards specified, conducting the beneficiation process or processes towards mineral recovery to the specification.

Scope

The scope covers the following :

- Recognize the ore processing requirements and related parameters
- Conduct leaching
- Conduct separation
- Conduct floatation, dewatering and filtration

Elements and Performance Criteria

Recognize the ore processing requirements and related parameters

To be competent, the user/individual on the job must be able to:

- PC1.** recognize the ore processing methodology and process to be adopted through discussions with the supervisor and reading the process manuals/ work instructions/standard operating procedures
- PC2.** discuss various reagents and their quantities to be used as well as effective reagents handling techniques
- PC3.** monitor the reagent addition rate and mix tank levels continuously during the ore processing
- PC4.** identify the various tools to be used at different steps of ore processing like hand & power tools, pneumatically & hydraulically powered tools, scaffolds, ladders etc.
- PC5.** perform the operations of various equipment used at various stages like mobile equipment and lifting equipment
- PC6.** identify the operations for various tools & equipment as well as the effective equipment handling guidelines
- PC7.** prepare for the emergency situations with required instructions and equipment in place

Conduct leaching

To be competent, the user/individual on the job must be able to:

- PC8.** conduct checks on various operational parameters like tank level, ph level, reagent usage, air parameters, alarms, pumps, screens, monitors etc.
- PC9.** add reagents as required, such as acids, carbon, cyanide, calcium as per the process requirements
- PC10.** conduct leaching/ ore beneficiation process to extract the minerals from the grinded ores
- PC11.** check and monitor threshold limit values (tlvs)
- PC12.** guard all identified hazards using rope/barriers and/or signs, clean up spills and leaks, report condition to appropriate personnel

Conduct separation

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To be competent, the user/individual on the job must be able to:

- PC13.** check various separation systems like drive system, drums, concentrators, jigs, classifiers etc.
- PC14.** check various operating parameters like operating levels, pressures, cleanliness
- PC15.** ensure the maintenance of flow rate
- PC16.** check and monitor threshold limit values (tlvs)
- PC17.** guard all identified hazards using rope/barriers and/or signs, clean up spills and leaks, report condition to appropriate personnel

Conduct floatation, dewatering and filtration

To be competent, the user/individual on the job must be able to:

- PC18.** identify the different types of floatation equipment including floatation air, bank of cells, conditioner, collector, floatation column, cells, such as: roughers, scavenger, cleaners, pumping system in-line assay system
- PC19.** start the floatation system and ensure adequate flow
- PC20.** control the quantity of reagents, percent solid, addition of air, level of froth/pulp, launder water, pump speed, ph level
- PC21.** stop floatation circuit in sequence
- PC22.** ensure that the reagent systems contain proper amount, type and strength of reagents
- PC23.** identify different types of dewatering systems including stock tanks, decanters, pumping systems, sampling systems, rakes, drive system, feed wells etc.
- PC24.** operate and maintain dewatering system while checking the shell, rakes (e.g. height, integrity), motor and gear box for unusual noises, pumps, water supply, pumps and valves etc.
- PC25.** guard all identified hazards using rope/barriers and/or signs, clean up spills and leaks, report condition to appropriate personnel
- PC26.** identify the different types of filtration systems and components including stock tanks, pumping systems, sampling systems, drive systems, lubrication systems, hydraulic systems, pneumatic systems, vacuum systems, filter cloth
- PC27.** check the functioning of vacuum system, gear box, bearings etc.
- PC28.** check the condition of the filter (e.g. thickness of filter cake)
- PC29.** conduct filtration while controlling the moisture, filtrate, wash water, pump speed, pressure, density etc.
- PC30.** monitor boot levels and agitation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** the hierarchy for reporting identified problems
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process

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- KU6.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to mineral processing operations
- KU7.** different types of mines and detail of the mine one is working in
- KU8.** benching in quarries, dressing of overhangs, undercuts, fencing
- KU9.** importance of first aid and hygiene
- KU10.** code of practice in specific areas of the mine
- KU11.** standing orders in force at the mine
- KU12.** importance of safety in the vicinity of machinery
- KU13.** about shot-firing / blasting related safety regulations including taking shelter during blasting
- KU14.** duties of workmen under the Mines Act-1952
- KU15.** provision of compensation and working hours, leaves, etc. as per Mines Act 1952
- KU16.** the outcome of violation of safety procedures
- KU17.** emergency response /disaster management plan prepared by the organization
- KU18.** different types of mineral processing processes and associated equipment. hazards and safety aspects involved in ore processing activities, handling reagents and usage of relevant PPEs
- KU19.** various separation systems like drive system, drums, concentrators, jigs, classifiers etc.
- KU20.** reagents such as acids, carbon, cyanide, calcium as per the process requirements
- KU21.** different types of floatation equipment including floatation air, bank of cells, conditioner, collector, floatation column, cells, such as: roughers, scavenger, cleaners, pumping system, in-line assay system, scrubbers etc.
- KU22.** different types of dewatering systems include: stock tanks, decanters, pumping systems, sampling systems, rakes, drive system, feed wells
- KU23.** equipment like shell, rakes (e.g. height, integrity), motor and gear box for unusual noises, U/F pumps, etc.
- KU24.** different types of filtration systems and components including stock tanks, pumping systems, sampling systems, drive systems, lubrication systems, hydraulic systems, pneumatic systems, vacuum systems, filter cloth
- KU25.** vacuum system, gear box, bearings, PLC, RLC operation (Logic controls), Pump, hydraulic systems, motors, valves etc.
- KU26.** hazards and safety aspects involved in ore processing activities and usage of relevant PPEs

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** discuss task lists, schedules and activities
- GS5.** enter the information in online Enterprise resource planning (ERP) systems under guidance of the supervisor



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- GS6.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS7.** make decisions pertaining to the concerned area of work
- GS8.** plan and organize the work order and tasks
- GS9.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS10.** detect problems in day to day tasks
- GS11.** discuss possible solution with the supervisor for problem solving
- GS12.** make decisions in emergency conditions
- GS13.** follow instructions and work on areas of improvement identified
- GS14.** complete the assigned tasks timely
- GS15.** use reasoning skills to identify and resolve basic problems
- GS16.** analyze and detect any potential problems which could arise during operation
- GS17.** respect persons with disabilities
- GS18.** explain the importance of gender-sensitization at work site
- GS19.** state basic laws, acts and provisions defined for Persons with Disability (PwD) by the statutory bodies

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Recognize the ore processing requirements and related parameters</i>	7	14	-	4
PC1. recognize the ore processing methodology and process to be adopted through discussions with the supervisor and reading the process manuals/ work instructions/standard operating procedures	1	2	-	1
PC2. discuss various reagents and their quantities to be used as well as effective reagents handling techniques	1	2	-	1
PC3. monitor the reagent addition rate and mix tank levels continuously during the ore processing	1	2	-	1
PC4. identify the various tools to be used at different steps of ore processing like hand & power tools, pneumatically & hydraulically powered tools, scaffolds, ladders etc.	1	2	-	1
PC5. perform the operations of various equipment used at various stages like mobile equipment and lifting equipment	1	2	-	-
PC6. identify the operations for various tools & equipment as well as the effective equipment handling guidelines	1	2	-	-
PC7. prepare for the emergency situations with required instructions and equipment in place	1	2	-	-
<i>Conduct leaching</i>	5	10	-	3
PC8. conduct checks on various operational parameters like tank level, ph level, reagent usage, air parameters, alarms, pumps, screens, monitors etc.	1	2	-	1
PC9. add reagents as required, such as acids, carbon, cyanide, calcium as per the process requirements	1	2	-	1
PC10. conduct leaching/ ore beneficiation process to extract the minerals from the grinded ores	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. check and monitor threshold limit values (tlvs)	1	2	-	-
PC12. guard all identified hazards using rope/barriers and/or signs, clean up spills and leaks, report condition to appropriate personnel	1	2	-	-
<i>Conduct separation</i>	5	6	-	4
PC13. check various separation systems like drive system, drums, concentrators, jigs, classifiers etc.	1	2	-	-
PC14. check various operating parameters like operating levels, pressures, cleanliness	1	1	-	1
PC15. ensure the maintenance of flow rate	1	1	-	1
PC16. check and monitor threshold limit values (tlvs)	1	1	-	1
PC17. guard all identified hazards using rope/barriers and/or signs, clean up spills and leaks, report condition to appropriate personnel	1	1	-	1
<i>Conduct floatation, dewatering and filtration</i>	13	20	-	9
PC18. identify the different types of floatation equipment including floatation air, bank of cells, conditioner, collector, floatation column, cells, such as: roughers, scavenger, cleaners, pumping system in-line assay system	1	2	-	-
PC19. start the floatation system and ensure adequate flow	1	2	-	1
PC20. control the quantity of reagents, percent solid, addition of air, level of froth/pulp, launder water, pump speed, ph level	1	2	-	1
PC21. stop floatation circuit in sequence	1	2	-	1
PC22. ensure that the reagent systems contain proper amount, type and strength of reagents	1	2	-	1
PC23. identify different types of dewatering systems including stock tanks, decanters, pumping systems, sampling systems, rakes, drive system, feed wells etc.	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. operate and maintain dewatering system while checking the shell, rakes (e.g. height, integrity), motor and gear box for unusual noises, pumps, water supply, pumps and valves etc.	1	2	-	-
PC25. guard all identified hazards using rope/barriers and/or signs, clean up spills and leaks, report condition to appropriate personnel	1	1	-	-
PC26. identify the different types of filtration systems and components including stock tanks, pumping systems, sampling systems, drive systems, lubrication systems, hydraulic systems, pneumatic systems, vacuum systems, filter cloth	1	1	-	1
PC27. check the functioning of vacuum system, gear box, bearings etc.	1	1	-	1
PC28. check the condition of the filter (e.g. thickness of filter cake)	1	1	-	1
PC29. conduct filtration while controlling the moisture, filtrate, wash water, pump speed, pressure, density etc.	1	1	-	1
PC30. monitor boot levels and agitation	1	1	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4105
NOS Name	Beneficiation and Mineral recovery
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	2
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

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MIN/N4106: Tailing Management

Description

This OS unit is about treating the water and managing the tailings post filtration.

Scope

The scope covers the following :

- Treat the water
- Manage the tailings

Elements and Performance Criteria

Treat the water

To be competent, the user/individual on the job must be able to:

- PC1.** obtain water from the mineral recovery unit and send the same to the treatment area
- PC2.** treat water for chemicals and purify the same to make it fit for reclamation purposes
- PC3.** send water to reclamation area (settling ponds)
- PC4.** sample water and monitor levels of ponds

Manage the tailings

To be competent, the user/individual on the job must be able to:

- PC5.** treat the tailings to reduce the toxin levels
- PC6.** check the ph level
- PC7.** dispose of tailings to the appropriate area

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** the hierarchy for reporting identified problems
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process
- KU6.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to mineral processing operations
- KU7.** different types of mines and detail of the mine one is working in
- KU8.** benching in quarries, dressing of overhangs, undercuts, fencing
- KU9.** importance of first aid and hygiene
- KU10.** code of practice in specific areas of the mine
- KU11.** standing orders in force at the mine

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- KU12.** importance of safety in the vicinity of machinery
- KU13.** about shot-firing / blasting related safety regulations including taking shelter during blasting
- KU14.** duties of workmen under the Mines Act-1952
- KU15.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU16.** the outcome of violation of safety procedures
- KU17.** emergency response /disaster management plan prepared by the organization
- KU18.** hazards and safety aspects involved in ore processing activities and usage of relevant pps
- KU19.** different types of ore processing processes and associated equipment
- KU20.** various water purifying agents
- KU21.** various waste treatment agents
- KU22.** hazards and safety aspects involved in ore processing activities and usage of relevant pps

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** discuss task lists, schedules and activities
- GS5.** enter the information in online Enterprise resource planning (ERP) systems under guidance of the supervisor
- GS6.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS7.** make decisions pertaining to the concerned area of work
- GS8.** plan and organize the work order and tasks
- GS9.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS10.** detect problems in day to day tasks
- GS11.** discuss possible solution with the supervisor for problem solving
- GS12.** make decisions in emergency conditions
- GS13.** follow instructions and work on areas of improvement identified
- GS14.** complete the assigned tasks timely
- GS15.** use reasoning skills to identify and resolve basic problems
- GS16.** analyze and detect any potential problems which could arise during operation
- GS17.** respect persons with disabilities
- GS18.** explain the importance of gender-sensitization at work site
- GS19.** state basic laws, acts and provisions defined for Persons with Disability (PwD) by the statutory bodies

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Treat the water</i>	17	29	-	11
PC1. obtain water from the mineral recovery unit and send the same to the treatment area	4	7	-	3
PC2. treat water for chemicals and purify the same to make it fit for reclamation purposes	4	8	-	2
PC3. send water to reclamation area (settling ponds)	5	7	-	3
PC4. sample water and monitor levels of ponds	4	7	-	3
<i>Manage the tailings</i>	13	21	-	9
PC5. treat the tailings to reduce the toxin levels	4	7	-	3
PC6. check the ph level	5	7	-	3
PC7. dispose of tailings to the appropriate area	4	7	-	3
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N4106
NOS Name	Tailing Management
Sector	Mining
Sub-Sector	Mineral Beneficiation
Occupation	Ore Processing
NSQF Level	4
Credits	2
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N4101.Pre-Operation checks of crusher and crusher sites	30	50	0	20	100	20
MIN/N4102.Operate conveyors and feeders	30	50	0	20	100	30
MIN/N4103.Operation and maintenance of the crusher	30	50	0	20	100	20
MIN/N1703.Follow Health, Safety, and Environmental Guidelines for opencast mines	30	50	-	20	100	20
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	140	230	-	80	450	100

Optional: 1 Grinding

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N4104.Grinding of crushed ore/mineral	30	50	0	20	100	20
Total	30	50	-	20	100	20

Optional: 2 Beneficiation



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N4105.Beneficiation and Mineral recovery	30	50	0	20	100	20
Total	30	50	-	20	100	20

Optional: 3 Tailing

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N4106.Tailing Management	30	50	0	20	100	20
Total	30	50	-	20	100	20

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
QP	Qualification Pack
NSQF	National Skills Qualifications Framework
TVET	Technical and Vocational Education and Training
SCMS	Skill Council for Mining Sector
ROPS	Roll-Over Protection Structure
FOPS	Falling Object Protective Structure
PC	Performance Criteria
HEMM	Heavy Earth Moving Machinery
OEM	Original Equipment Manufacturer
DGMS	Directorate General of Mining Safety
MMR	Metalliferous Mines Regulations
CMR	Coal Mines Regulations
SOP	Standard Operating Practice
MVTR	Mine Vocational Training Rules
MRR	Mines Rescue Rules
HSE	health, safety & environment
OB	Over Burden
ERP	Enterprise resource planning
MCDR	Mineral Conservation and Development Rules

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	A sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	a sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in the industry.
Occupational Standards	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.

Qualification Pack

Qualification Pack	QP comprises the set of OS, together with the educational, training, and other criteria required to perform a job role. A QP is assigned a unique qualifications pack role
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.